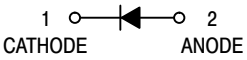
# Switching Diode

# MMSD914, SMMSD914



**SOD−123**
**CASE 425**
**PLASTIC**




### Features

- SOD−123 Surface Mount Package
- High Breakdown Voltage
- Fast Speed Switching Time
- S Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable
- These Devices are Pb−Free, Halogen Free/BFR Free and are RoHS Compliant*

### MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Continuous Reverse Voltage | $V_R$ | 100 | Vdc |
| Peak Forward Current | $I_F$ | 200 | mAdc |
| Peak Forward Surge Current | $I_{FM(surge)}$ | 500 | mAdc |
| Non−repetitive Peak Forward Surge Current<br>Pulse Width =1 second<br>Pulse Width =1 micro second | $I_{FSM}$ | <br>1.0<br>2.0 | <br>A<br>A |

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

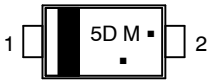
### THERMAL CHARACTERISTICS

| Characteristic | Symbol | Max | Unit |
|---|---|---|---|
| Total Device Dissipation FR−5 Board (Note 1)<br>$T_A$ = 25°C<br>Derate above 25°C | $P_D$ | 425<br><br>3.4 | mW<br><br>mW/°C |
| Thermal Resistance, Junction−to−Ambient | $R_{\theta JA}$ | 290 | °C/W |
| Junction and Storage Temperature Range | $T_J$, $T_{stg}$ | −55 to +150 | °C |

1. FR−5 = 1.0oz Cu, 1.0in$^2$ pad

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

### MARKING DIAGRAM




5D = Specific Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

### ORDERING INFORMATION

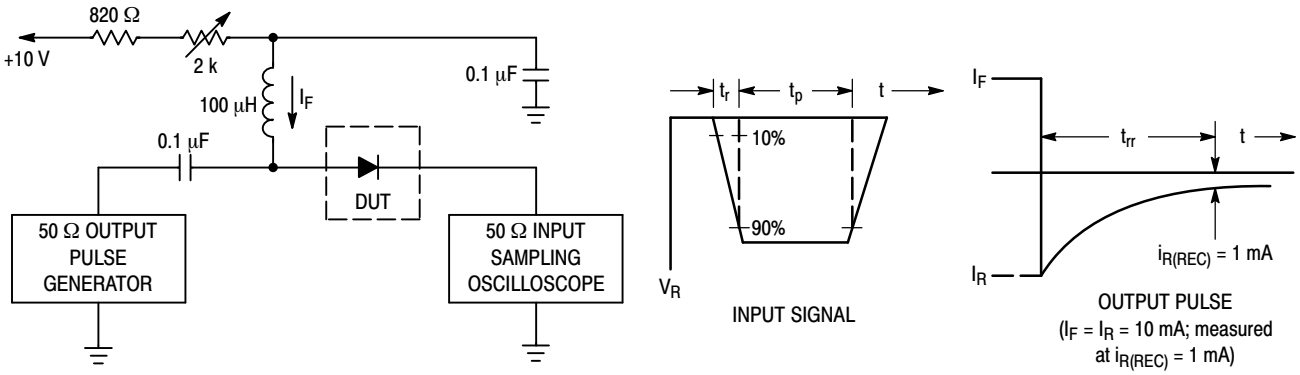
| Device | Package | Shipping† |
|---|---|---|
| MMSD914T1G | SOD−123 (Pb−Free) | 3,000 / Tape & Reel |
| SMMSD914T1G | SOD−123 (Pb−Free) | 3,000 / Tape & Reel |
| MMSD914T3G | SOD−123 (Pb−Free) | 10,000 / Tape & Reel |
| SMMSD914T3G | SOD−123 (Pb−Free) | 10,000 / Tape & Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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**ELECTRICAL CHARACTERISTICS** ($T_A$ = 25°C unless otherwise noted)

| Characteristic | Symbol | Min | Max | Unit |
|---|---|---|---|---|
| **OFF CHARACTERISTICS** | | | | |
| Reverse Breakdown Voltage ($I_{BR}$ = 100 μAdc) | $V_{(BR)}$ | 100 | – | Vdc |
| Reverse Voltage Leakage Current<br>($V_R$ = 20 Vdc)<br>($V_R$ = 75 Vdc) | $I_R$ | <br>–<br>– | <br>25<br>5.0 | <br>nAdc<br>μAdc |
| Forward Voltage ($I_F$ = 10 mAdc) | $V_F$ | – | 1000 | mVdc |
| Diode Capacitance ($V_R$ = 0 Vdc, f = 1.0 MHz) | $C_D$ | – | 4.0 | pF |
| Reverse Recovery Time ($I_F$ = $I_R$ = 10 mAdc) (Figure 1) | $t_{rr}$ | – | 4.0 | ns |




Notes: 1. A 2.0 kΩ variable resistor adjusted for a Forward Current ($I_F$) of 10 mA.
2. Input pulse is adjusted so $I_{R(peak)}$ is equal to 10 mA.
3. $t_p \gg t_{rr}$

**Figure 1. Recovery Time Equivalent Test Circuit**



Share Feedback
Your Opinion Matters

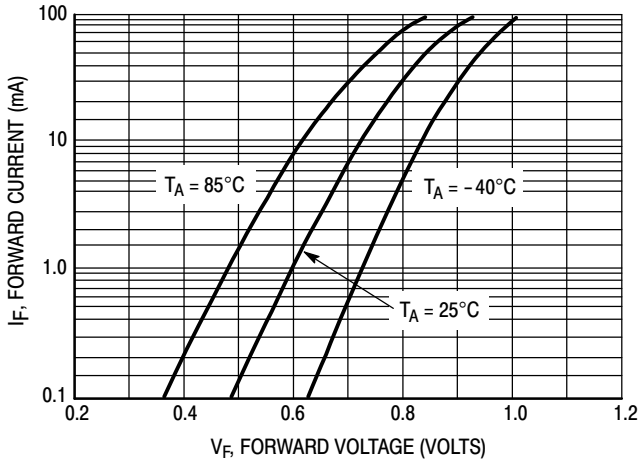



**Figure 2. Forward Voltage**

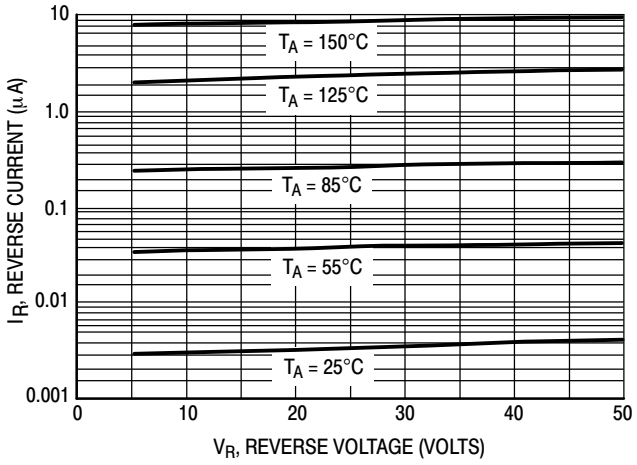



**Figure 3. Leakage Current**

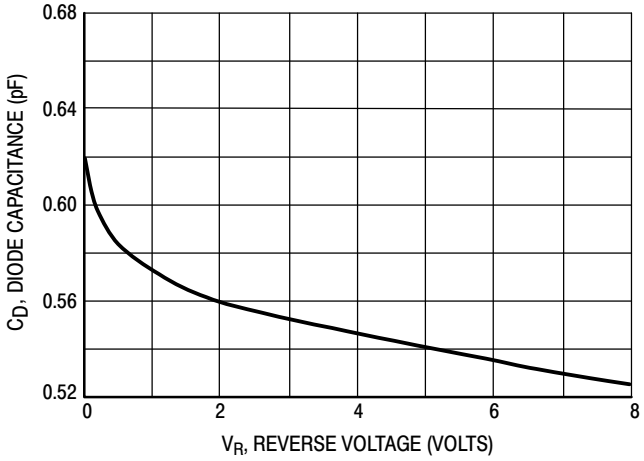
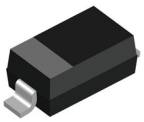



**Figure 4. Capacitance**

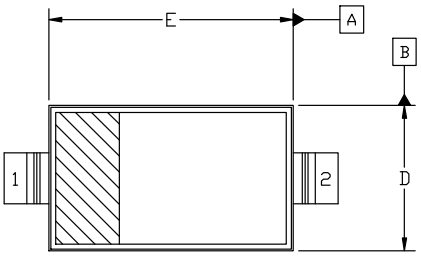


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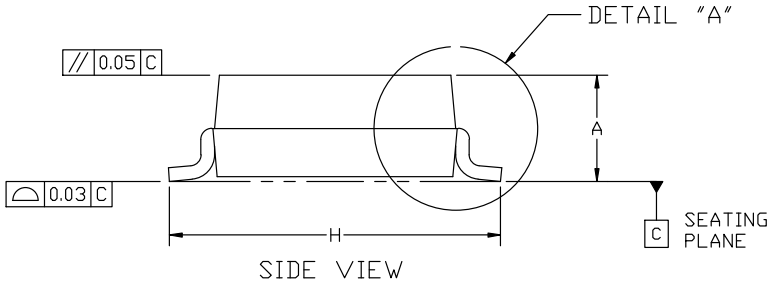


**SOD−123 2−LEAD, 1.60x2.69x1.16**
CASE 425
ISSUE H

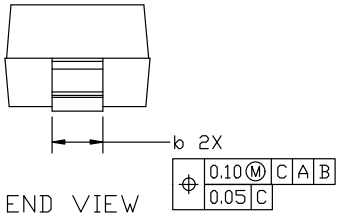
DATE 29 FEB 2024



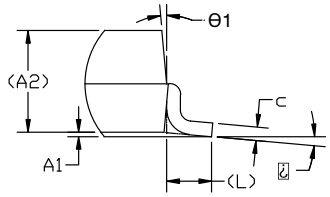

TOP VIEW




SIDE VIEW



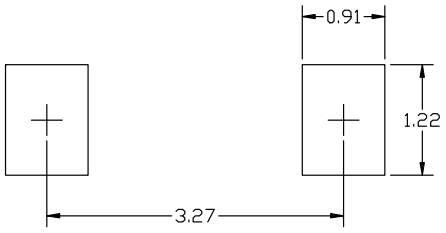

END VIEW




DETAIL "A"

NOTES:
1. DIMENSION AND TOLERANCING PER ASME Y14.5M, 2018
2. CONTROLLING DIMENSION: MILLIMETERS

| | MILLIMETER | | |
|---|---|---|---|
| DIM | MIN. | NOM. | MAX. |
| A | 0.94 | 1.17 | 1.35 |
| A1 | 0.00 | 0.05 | 0.10 |
| A2 | | 1.16 REF. | |
| b | 0.51 | 0.61 | 0.71 |
| c | – | – | 0.15 |
| D | 1.40 | 1.60 | 1.80 |
| E | 2.54 | 2.69 | 2.84 |
| H | 3.56 | 3.68 | 3.86 |
| L | | 0.25 REF. | |
| ∠ | 0° | | 10° |
| θ1 | 0° | | 10° |




RECOMMENDED MOUNTING FOOTPRINT

*For additional information on or Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference manual SOLDERRM/D.

## GENERIC MARKING DIAGRAM*




XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package
(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

STYLE 1:
PIN 1. CATHODE
2. ANODE

| | | |
|---|---|---|
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| **DESCRIPTION:** | **SOD−123 2−LEAD, 1.60x2.69x1.16** | **PAGE 1 OF 1** |

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